PROGRAM CODE:

% Calculation and Verification of Frequency Deviation % and Modulation Index of FM Signal

clc; clear; close all;

% Message Signal Parameters Am = 2; fm = 500;

% Carrier Signal Parameters Ac = 1; fc = 10000;

% FM Parameters kf = 250; fs = 100000; T = 0.01;

% Time Vector t = 0:1/fs:T;

% Message Signal m = Am*cos(2*pi*fm*t);

% Frequency Deviation delta_f = kf*Am;

% Modulation Index beta = delta_f/fm;

% FM Signal fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));

% Display Parameters fprintf('Carrier Frequency = %.0f Hz\n',fc); fprintf('Message Frequency = %.0f Hz\n',fm); fprintf('Frequency Deviation = %.2f Hz\n',delta_f); fprintf('Modulation Index = %.2f\n',beta);

% Verification beta_verify = delta_f/fm;

fprintf('Verified Modulation Index = %.2f\n',beta_verify);

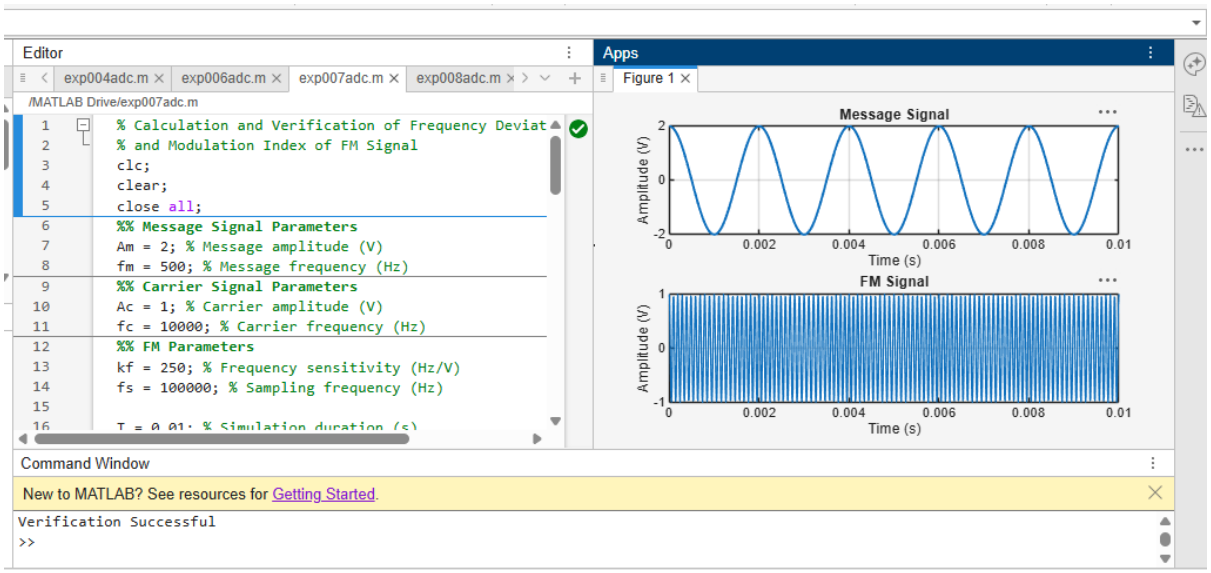
if abs(beta-beta_verify) < 1e-6 fprintf('Verification Successful\n'); else fprintf('Verification Failed\n'); end

% Plotting figure;

subplot(2,1,1); plot(t,m,'LineWidth',1.5); grid on; title('Message Signal'); xlabel('Time (s)'); ylabel('Amplitude (V)');

subplot(2,1,2); plot(t,fm_signal,'LineWidth',1); grid on; title('FM Signal'); xlabel('Time (s)'); ylabel('Amplitude (V)');

OUTPUT:


Editor
exp004adc.m
exp006adc.m
exp007adc.m
exp008adc.m
/MATLAB Drive/exp007adc.m
% Calculation and Verification of Frequency Deviat
% and Modulation Index of FM Signal
clc;
clear;
close all;
%% Message Signal Parameters
Am = 2; % Message amplitude (V)
fm = 500; % Message frequency (Hz)
%% Carrier Signal Parameters
Ac = 1; % Carrier amplitude (V)
fc = 10000; % Carrier frequency (Hz)
%% FM Parameters
kf = 250; % Frequency sensitivity (Hz/V)
fs = 100000; % Sampling frequency (Hz)
Apps
Figure 1
Message Signal
Amplitude (V)
Time (s)
FM Signal
Amplitude (V)
Time (s)
Command Window
New to MATLAB? See resources for Getting Started.
Verification Successful
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